

2(a)(i)	<i>closed circulation because</i> blood is contained within blood vessels ; <i>double circulation because</i> blood flows through the heart twice for one complete, circuit of the body / circulation ; A described A pulmonary circulation and systemic circulation	2
2(a)(ii)	Q = vena cava and R = aorta ;	1

Question	Answer	Marks
2(a)(iii)	I ref to valves any three from: Q has a: thinner wall ; A thinner middle layer wider lumen relative to thickness of wall ; thinner tunica media ; (proportionately) less smooth muscle ; (proportionately) less / very few, elastic fibres ; allow ecf from Q2(a)(ii)	3
2(b)(i)	human antibodies, are made of four polypeptides / contain two light chains and two heavy chains ;	1
2(b)(ii)	any two from: more / easier to, pass through gaps in, capillary wall / endothelium ; (so) more enter tissue fluid (surrounding tumour cells) ; <i>idea of</i> more cancer cells destroyed quickly ; (smaller so) may not trigger an immune response ; AVP ; e.g. easier to, diffuse / move, through tumour to reach, more / all, cells binds more tightly to antigens on (cancer) cell surface(s) <i>suggestion that</i> makes macrophage response stronger smaller overall size even when drugs attached	2

Question	Answer	Marks
2(b)(iii)	any three from: gives (artificial) passive immunity ; fast acting / quick response / time not needed for immune response ; antibody binds to (non-self / foreign) antigen (on surface of a pathogen) ; antibodies bind to toxins, neutralising them / AW or antibodies act as antitoxins ; ref phagocytosis ; AVP ; e.g. opsonisation bind to flagella and immobilise pathogen antibodies cause, agglutination / clumping / AW, of pathogen	3